
User Adoption Mechanisms of Autonomous Ride-Hailing in Smart Mobility

Contexts: An Empirical Analysis Based on the Technology Acceptance Model and Trust Theory

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Abstract: With the accelerating commercialization of autonomous driving technologies, autonomous ride-hailing services are gradually becoming an important carrier of smart mobility. However, insufficient user trust and unstable adoption intention remain key barriers to their market diffusion. Drawing on the Technology Acceptance Model (TAM) and trust theory, this study develops an integrated cognition-trust-behavioral intention framework to systematically examine how perceived usefulness, perceived ease of use, and trust shape users' willingness to adopt autonomous ride-hailing services. Based on 633 valid questionnaires, the study employs partial least squares structural equation modeling (PLS-SEM) for empirical testing. The results show that both perceived usefulness and perceived ease of use significantly and positively affect user trust and usage intention; trust plays a crucial mediating role between cognitive evaluations and adoption intention; and perceived usefulness exerts a stronger effect on trust and behavioral intention than perceived ease of use. The study extends the explanatory boundary of TAM in autonomous driving contexts and, in practical terms, proposes a trust-centered pathway for promoting user adoption.

Keywords: autonomous ride-hailing; Technology Acceptance Model; trust; PLS-SEM; user adoption

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Received 15 February 2026, Revised 01 March 2026, Accepted 25 March 2026

1 Introduction

1.1 Research Background and Significance

With the rapid development of artificial intelligence, vehicle-to-everything technologies, and advanced autonomous driving, the traditional ride-hailing industry is undergoing a profound transition from human-driven services to intelligence-driven services. Compared with conventional ride-hailing modes that rely on human drivers, autonomous ride-hailing offers clear advantages in reducing labor costs, improving vehicle utilization, optimizing the allocation of spatiotemporal transport resources, and enhancing the scalability of platform operations. For this reason, autonomous mobility services have become a major direction in the global convergence of intelligent transportation and the platform economy.

In recent years, the commercialization of autonomous driving has accelerated continuously. Companies such as Waymo, Zoox, and Apollo Go have launched pilot or regular robotaxi operations in selected cities, indicating that autonomous driving is moving from laboratory testing to deployment in real traffic environments. In China in particular, the state has steadily refined the legal and regulatory framework for autonomous driving and has promoted implementation through pilot projects and demonstration zones. Apollo Go, for example, has achieved regularized operations in cities such as Beijing, Wuhan, and Chengdu, and has reached a relatively large-scale deployment in Wuhan, suggesting that the commercialization of autonomous driving already has a meaningful practical foundation.

From a comparative international perspective, the United States has developed through a pattern of enterprise-driven innovation, local pilot expansion, and federal regulatory guidance; China has followed a pathway that combines policy support, scenario openness, and platform expansion; and South Korea and Japan have tended to advance autonomous driving from the perspective of public transportation and urban governance. This indicates that autonomous driving is no longer a purely technical issue. Rather, it has become an important variable in the restructuring of transportation systems, the optimization of urban governance, and the upgrading of platform competition.

Yet technical feasibility and commercial feasibility do not automatically translate into user acceptability. While autonomous ride-hailing promises efficiency gains and business-model innovation, its algorithmic decision-making, autonomous control, and ambiguous allocation of responsibility continue to generate public concerns regarding safety, reliability, and controllability. In other words, the central bottleneck in the diffusion of autonomous ride-hailing is no longer simply technological maturity, but whether users are willing to trust and adopt this new mobility mode.

Accordingly, a systematic examination of the adoption mechanism of autonomous ride-hailing from the perspective of user behavior is not only of substantial practical importance, but also provides theoretical support for platform strategy optimization, regulatory improvement, and the broader social transition toward intelligent mobility.

Existing research on ride-hailing and smart mobility has generated a rich theoretical foundation. Early studies mainly focused on supply-demand matching, routing and dispatching, dynamic pricing, and market equilibrium in taxi and ride-hailing markets, explaining how platforms improve resource allocation through prices, algorithms, and organizational mechanisms. As the platform economy and smart mobility evolved, the literature expanded to driver supply structures, platform governance, peak-hour dispatching strategies, service efficiency, and congestion effects. With the introduction of autonomous driving, the research agenda has further extended from platform operational efficiency to technological deployment, ethical risk, institutional constraints, and social acceptance. Existing studies generally suggest that autonomous driving may reshape urban mobility, yet its large-scale diffusion remains constrained by risk perception, institutional trust, and public acceptance. Especially in road-traffic contexts, the autonomous decision-making character of these systems means that users cannot rely solely on functional judgments when deciding whether to adopt them; they must also evaluate safety risks, system reliability, and the consequences of responsibility allocation.

At the user level, the Technology Acceptance Model (TAM) provides a classic framework for explaining new technology adoption. TAM argues that perceived usefulness and perceived ease of use are the core determinants of behavioral intention. However, autonomous ride-hailing differs from general information technologies or platform tools because its use involves

substantially higher uncertainty and potential risk. As a result, TAM alone is insufficient to fully explain adoption logic in autonomous driving contexts.

Trust theory offers an important complement. In high-uncertainty technological environments, trust not only helps users reduce perceived risk, but also serves as a crucial bridge through which cognitive evaluations are translated into behavioral decisions. Prior studies show that user trust in autonomous driving is affected by technical performance, information transparency, human-machine interaction, and safety incidents, and that it ultimately shapes whether users are willing to try, accept, and continue using related services.

Taken together, the existing literature has explored platform mechanisms, technology applications, and user behavior from different angles, but it still lacks a systematic integration of the internal path linking cognition, trust, and behavioral intention in the case of autonomous ride-hailing. Especially in a high-risk technological setting such as autonomous driving, how TAM and trust theory can be combined into a more explanatory adoption model remains an important question.

1.2 Research Questions

As autonomous ride-hailing moves toward commercialization, public attention to the technology continues to increase. Compared with traditional mobility services, users evaluating autonomous ride-hailing not only consider whether it is efficient and convenient, but also whether it is safe, credible, and worthy of reliance. User adoption is therefore no longer a purely instrumental choice; it is a composite decision process that integrates technological cognition, risk judgment, and psychological trust.

Although previous studies have addressed acceptance of autonomous driving, three gaps remain. First, most studies focus on technological feasibility, regulation, or platform operating efficiency, while the micro-level mechanisms of user adoption are relatively underexplored. Second, even when TAM is used to explain new technology adoption, trust mechanisms in high-uncertainty contexts often receive insufficient attention, which limits explanation of the full logic of intention formation. Third, the relationships among perceived usefulness, perceived ease of use, trust, and usage intention in the autonomous ride-hailing context still lack systematic empirical verification.

Accordingly, taking the Apollo Go platform as the focal case, this study integrates TAM and trust theory to address four questions:

First, how do perceived usefulness and perceived ease of use affect users' trust in autonomous ride-hailing services?

Second, does trust mediate the relationship between users' cognitive evaluations and their usage intention?

Third, do perceived usefulness and perceived ease of use differ in the strength of their effects on user adoption?

Fourth, based on the empirical results, how should platforms and regulators construct more effective mechanisms for fostering user trust?

1.3 Research Significance

1.3.1 Theoretical Significance

The theoretical significance of this study is reflected in three aspects.

First, the study advances the focus of research from technological feasibility and platform operating mechanisms to the micro-level process of user adoption, thereby broadening the analytical perspective of autonomous mobility research. Compared with prior studies that concentrate more heavily on system efficiency and business models, this study examines how users form trust and make adoption decisions, which more directly addresses a core theoretical issue in the diffusion of autonomous driving.

Second, the study introduces trust into the TAM framework and develops an integrated cognition-trust-behavioral intention model, thereby extending the explanatory boundary of TAM in high-uncertainty technological settings. This framework not only helps explain the adoption mechanism of autonomous ride-hailing, but also offers a potentially transferable theoretical path for studying user behavior toward other autonomous systems and high-risk intelligent technologies.

Third, by using structural equation modeling to test the relationships among variables, the study clarifies the mediating

role of trust between cognitive variables and behavioral intention, thereby deepening theoretical understanding of the adoption mechanism of autonomous driving technologies.

1.3.2 Practical Significance

The practical significance of this study is reflected in several aspects.

First, the study helps identify the key factors influencing user adoption of autonomous ride-hailing and thus provides an empirical basis for optimizing service design and user communication strategies. In particular, in a context where public safety concerns remain salient, understanding the mechanism of trust formation has direct practical value for expanding market acceptance.

Second, the findings provide concrete implications for platform operation and product improvement. Platforms should attach greater importance to algorithmic transparency, safety feedback mechanisms, and user-experience optimization in order to enhance users' understanding, predictability, and acceptance of the system.

Third, the study also offers a reference for governments seeking to improve autonomous-driving regulation. As commercialization enters an accelerated phase, regulatory priorities are shifting from pure testing management to responsibility attribution, data security, information disclosure, and the governance of social trust. The empirical analysis presented here provides more targeted support for related institutional design.

1.4 Research Content and Technical Roadmap

1.4.1 Research Object and Content

This study focuses on autonomous ride-hailing services and uses the Apollo Go platform as a representative case. It pays particular attention to the cognitive mechanism, trust formation, and usage intention of ordinary users when they encounter, understand, and evaluate such services. Around this research object, the study undertakes four major tasks.

First, by reviewing the literature on ride-hailing platforms, autonomous-driving adoption, and trust theory, the study identifies the theoretical foundation and research gaps in user adoption research on autonomous ride-hailing services.

Second, by integrating TAM and trust theory, it develops a theoretical model that includes perceived usefulness, perceived ease of use, trust, and usage intention, and then formulates the corresponding hypotheses.

Third, it collects sample data through a questionnaire survey and tests reliability and validity, structural paths, and mediation effects.

Fourth, based on the empirical results, it explains the internal logic of the user adoption mechanism of autonomous ride-hailing and proposes management implications and policy recommendations.

1.4.2 Technical Roadmap

Building on the existing literature and theoretical foundation, this study develops an analytical framework centered on TAM and trust theory and conducts a systematic investigation of user adoption of autonomous ride-hailing. The overall technical roadmap follows the logic of problem identification, theoretical construction, empirical testing, and result interpretation, and can be divided into five stages.

(1) Research problem identification and theoretical foundation construction

Against the background of accelerating commercialization of autonomous driving and insufficient user trust, the study first clarifies the core research problem of user adoption of autonomous ride-hailing. Through a systematic review of the literature on ride-hailing platforms, smart mobility, TAM, and trust theory, it builds the theoretical basis of the study and identifies the limitations of existing research in explaining user behavior under high uncertainty.

(2) Model construction and hypothesis development

On the basis of theoretical analysis, TAM and trust theory are integrated to construct a research model including perceived ease of use (PEOU), perceived usefulness (PU), trust, and usage intention (UI), thereby forming a cognition-trust-behavioral intention path. Corresponding hypotheses about the causal relationships among variables are then proposed.

(3) Questionnaire design and data collection

Using mature measurement scales and adapting them to the autonomous ride-hailing context, the study defines and

operationalizes each latent variable. A five-point Likert scale is used to measure perceived usefulness, perceived ease of use, trust, and usage intention. Valid sample data are collected through online channels from users who possess relevant cognition or potential willingness to use the service.

(4) Data analysis and model testing

After data preprocessing and descriptive statistical analysis, the study employs PLS-SEM to test the research model. Reliability and validity are first examined through Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE), with discriminant validity assessed using the Fornell-Larcker criterion and the HTMT ratio. Structural paths and hypotheses are then tested, mediation effects are examined with the bootstrap method, and model explanatory power (R^2) and effect sizes (f^2) are evaluated.

(5) Result interpretation and theoretical extension

Based on the empirical findings, the study systematically explains the paths through which perceived usefulness, perceived ease of use, and trust shape user adoption behavior, with particular attention to the mediating mechanism of trust and the differentiated effects of distinct cognitive variables. On this basis, the theoretical implications for extending TAM in high-uncertainty contexts are discussed, and management and policy recommendations for autonomous ride-hailing platforms are proposed.

In summary, the study develops a research path grounded in theoretical integration, supported by survey data, and centered on structural equation modeling. It thus establishes a complete analytical loop from theory construction to empirical testing and result interpretation, providing a systematic methodological framework for understanding the user adoption mechanism of autonomous ride-hailing.

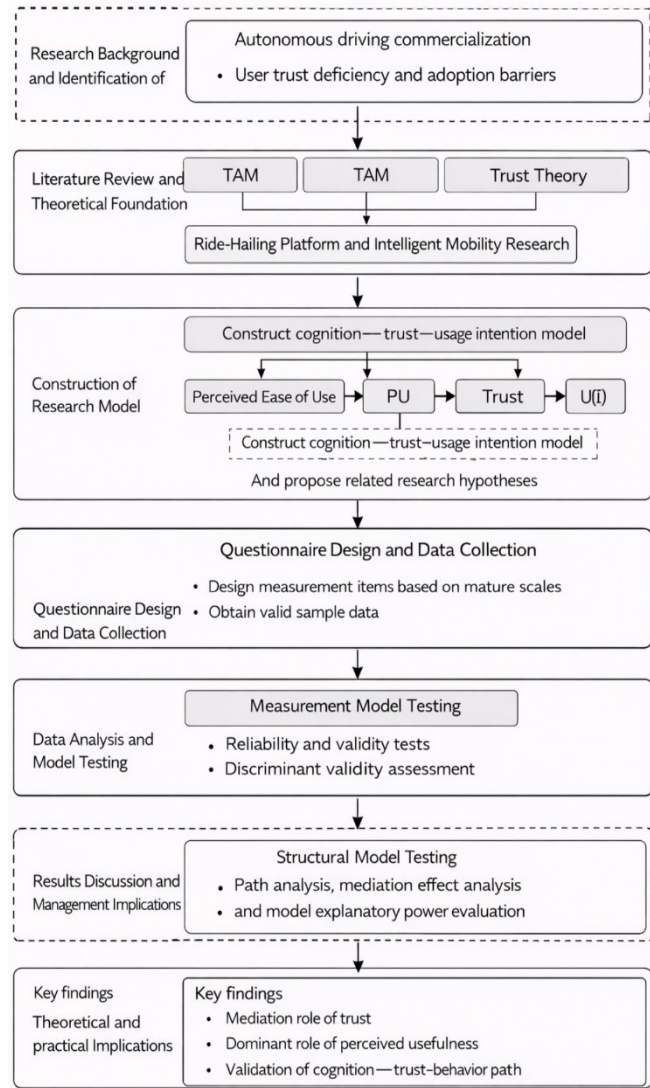


Figure 1. Research Framework and Technical Roadmap

1.5 Research Methods

This study adopts an empirical approach that combines questionnaire survey methods with structural equation modeling to systematically analyze the relationships among user cognition, trust, and behavioral intention in the context of autonomous ride-hailing. The overall design follows the analytical sequence of theoretical construction, variable measurement, and model testing so as to ensure both the reliability and explanatory power of the findings.

At the stage of theoretical construction, the study builds a research model based on the Technology Acceptance Model and trust theory, including perceived usefulness (PU), perceived ease of use (PEOU), trust, and usage intention (UI), and proposes corresponding hypotheses. The model is designed to reveal how users transform cognitive evaluations into trust formation and subsequently into behavioral decisions in a high-uncertainty technological environment.

In terms of variable measurement, the questionnaire design is based on established scales. The scales for perceived usefulness and perceived ease of use mainly draw on Venkatesh et al. (2012), while the trust scale draws on Gefen et al. (2003),

with contextual revisions made for autonomous ride-hailing to enhance measurement fit. All items are measured on a five-point Likert scale to ensure comparability and suitability for statistical analysis.

For data collection and sample processing, the survey was distributed online to users with relevant cognition or potential willingness to use autonomous ride-hailing. After invalid and abnormal cases were removed, 633 valid questionnaires were retained. To ensure data quality, descriptive statistics, normality checks, and outlier diagnostics were conducted before model testing.

For data analysis, the study employs partial least squares structural equation modeling (PLS-SEM). The measurement model is evaluated in terms of reliability and validity, including Cronbach's alpha, CR, and AVE, with discriminant validity assessed using the Fornell-Larcker criterion and the heterotrait-monotrait ratio (HTMT). The structural model is then tested by means of path analysis and bootstrap-based mediation analysis, while model explanatory power (R^2) and effect sizes (f^2) are also reported. Overall, the structural equation modeling approach enables a systematic explanation of the user adoption mechanism of autonomous ride-hailing.

1.6 Main Contributions

Focusing on the context of autonomous ride-hailing, this study constructs and validates an integrated analytical framework of user adoption based on TAM and trust theory. Its main contributions can be summarized at the methodological, theoretical, practical, and policy levels.

(1) Methodological contribution: a structural-equation-model-based analytical path for user adoption mechanisms

Compared with studies that rely on descriptive statistics or conventional regression alone, this study employs PLS-SEM to test the relationships among multiple latent variables and develops a complete empirical framework consisting of theoretical construction, scale measurement, path analysis, and mediation testing. This approach simultaneously handles complex latent structures and identifies both mediating mechanisms and path differences, thus strengthening explanatory depth and robustness.

(2) Theoretical contribution: extending TAM and strengthening the explanatory role of trust

By incorporating trust into the traditional TAM framework, the study constructs a cognition-trust-behavioral intention model and extends the explanatory boundary of TAM in high-uncertainty technological settings. The findings show that trust not only directly influences usage intention, but also mediates between cognitive variables and behavioral decisions, thereby revealing the multi-layered mechanism of user adoption in autonomous-driving scenarios.

(3) Practical contribution: proposing a trust-centered optimization path for autonomous mobility services

Based on the empirical evidence, the study identifies the key determinants of user adoption and proposes targeted optimization paths from the platform-operations perspective. Specifically, platforms should improve algorithmic transparency, strengthen safety-information communication, and optimize user experience in order to enhance comprehensibility, predictability, and trust, thereby promoting technology adoption.

(4) Policy and governance implication: building a trust-oriented regulatory logic for autonomous driving

The results suggest that stable user trust in autonomous driving cannot be established effectively through market mechanisms alone. Accordingly, trust governance should also be strengthened at the institutional level, including safety regulation, responsibility attribution, and information-disclosure transparency. This conclusion provides useful reference points for policy design and industry norm-setting in autonomous mobility.

2 Theoretical Foundation and Literature Review

2.1 Progress in Research on Autonomous Ride-Hailing

With the deep integration of the platform economy and autonomous-driving technologies, ride-hailing services are evolving from traditional human-driven modes toward intelligent and driverless modes. Existing studies have mainly addressed this topic from the perspectives of operating mechanisms, market structure, and technology application.

At the platform-operations level, early studies, grounded in equilibrium models of taxi markets, analyzed supply-demand

matching and resource-allocation efficiency (Yang & Wong, 1998). As ride-hailing platforms emerged, the literature gradually extended to issues such as dynamic pricing, ride pooling, and spatial heterogeneity (Zha et al., 2018). Scholars have further examined driver supply structures and platform organization, including the effects of multi-homing competition and labor mobility on service stability (Chen et al., 2024) as well as contract and sourcing strategies that optimize supply efficiency (Dong et al., 2024).

With the introduction of autonomous-driving technologies, the research focus has gradually shifted from operational optimization to technological shocks and industry restructuring. On the one hand, autonomous vehicles are expected to improve vehicle utilization substantially and reduce operating costs, thereby reshaping platform business models. On the other hand, their adoption also generates new social issues such as employment substitution and unequal resource distribution (Ao et al., 2024). Studies on mixed fleets further suggest that human-driven and autonomous vehicles may be complementary under different conditions, providing a pathway for gradual technological integration (Ao et al., 2024).

Although the above studies reveal the structural impact of autonomous driving on mobility systems at the macro level, they are still primarily supply-side and platform-oriented. Much less attention has been paid to the user-side mechanism of adoption. In particular, under the high-uncertainty conditions of autonomous ride-hailing, how users form cognitive evaluations and trust and how these are transformed into behavioral intention still lacks systematic analysis.

2.2 Application of TAM in the Intelligent Mobility Domain

The Technology Acceptance Model (TAM) is a classic framework for explaining user adoption behavior. Its core argument is that perceived usefulness (PU) and perceived ease of use (PEOU) are the key determinants of usage intention (Davis, 1989). TAM has been widely used in information systems and digital-platform research and has been continuously extended to new technological contexts.

Within mobility services, researchers have applied TAM to ride-hailing, shared mobility, and intelligent transportation systems, confirming the significant effects of PU and PEOU on user adoption intention. For example, users' perception that a service improves efficiency and reduces time costs tends to directly enhance their intention to use it, while ease of operation promotes acceptance by reducing learning costs.

However, the explanatory power of traditional TAM is challenged in autonomous-driving contexts. First, autonomous-driving technology is highly complex and black-boxed, making it difficult for users to directly evaluate its operating logic and thereby weakening the role of perceived ease of use. Second, autonomous systems involve personal safety risks and potentially irreversible decision outcomes, which means that users' decisions depend not only on functional cognition but also on risk and uncertainty. Consequently, PU and PEOU alone cannot adequately explain user adoption of autonomous-driving technologies.

Some studies have therefore introduced extended variables, such as perceived risk and social influence, to improve explanatory power. Even so, the systematic integration of trust remains insufficient, which constrains the theoretical extension of TAM in the field of autonomous driving.

2.3 Trust Theory and the Adoption of Autonomous Driving Technologies

In high-uncertainty technological environments, trust is widely regarded as a central psychological mechanism that shapes user behavior. Trust theory holds that when individuals face complex systems, they rely on subjective judgments about system reliability and safety to reduce perceived risk (Mayer et al., 1995). This mechanism is particularly critical in the autonomous-driving domain.

Existing studies indicate that users' trust in autonomous-driving systems is shaped not only by technological performance, but also by safety incidents, information transparency, and human-machine interaction. For instance, the system's ability to handle abnormal situations, the clarity of accident liability, and the explainability of algorithmic decisions all significantly affect trust levels. Negative experiences or safety incidents may also have long-lasting inhibitory effects on trust and thereby reduce adoption willingness.

Further research suggests that trust mediates the relationship between user cognition and behavior. On the one hand,

functional evaluations of a technology, such as efficiency and convenience, are translated into behavioral intention through trust. On the other hand, trust can also directly influence decision-making. In autonomous-driving contexts, trust is therefore not merely an independent variable, but a key bridge linking cognition and action.

Nevertheless, most existing studies analyze trust from a single perspective and rarely embed it systematically within a broader technology-acceptance framework. Moreover, the differentiated effects of trust across different cognitive variables and its underlying path mechanisms still require further empirical testing.

2.4 Literature Review and Hypothesis Development

In summary, existing studies have generated important insights into autonomous ride-hailing and user adoption, but several limitations remain.

First, from the perspective of research focus, existing studies pay more attention to platform operation and technological application than to the micro-level mechanism of user adoption.

Second, from the theoretical perspective, traditional TAM is not fully suited to high-risk and high-uncertainty contexts, and trust has not yet been systematically integrated.

Third, from the perspective of mechanism analysis, the path linking cognition, trust, and behavior still lacks a unified explanation.

To address these limitations, this study integrates TAM and trust theory to construct a theoretical model centering on perceived usefulness, perceived ease of use, trust, and usage intention. Special attention is paid to the mediating role of trust in the user-adoption process, and the related hypotheses are tested empirically.

3 Research Object, Data Sources, and Preliminary Analysis

3.1 Research Object and Data Sources

This study focuses on autonomous ride-hailing services and selects Apollo Go as a representative case. The choice of this platform is justified by two considerations. First, it has already achieved regularized operations in several Chinese cities and thus has strong practical representativeness. Second, the platform is at the forefront of the industry in terms of commercialization, scenario application, and public visibility, so it is well suited to reflecting the current developmental characteristics of autonomous ride-hailing and the state of user cognition.

During the preliminary research stage, the study combined publicly available online discussions with questionnaire-survey data to capture both the cognitive basis and the behavioral tendencies of the public toward autonomous ride-hailing. Online-discussion materials were used primarily to identify major public concerns and typical risk perceptions, whereas questionnaire data were used for subsequent empirical testing of the theoretical model. With respect to online materials, the study focused on review videos and comment sections on the Bilibili platform related to Apollo Go. These comments often revolve around technical performance, riding experience, traffic safety, responsibility attribution, and industry prospects, and thus concentratedly reflect public views on autonomous ride-hailing in a naturalistic discourse environment.

During data organization, the relevant comments were filtered and summarized to generate a foundational understanding of major public concerns. Overall, online discussions provided important real-world references for subsequent questionnaire design and variable specification, allowing the study to be embedded more deeply in the actual context of autonomous-driving commercialization.

3.2 Preliminary Identification of Public Concerns

The preliminary materials indicate that public discussion of autonomous ride-hailing does not focus narrowly on technology itself; rather, it exhibits a multidimensional structure that is mainly concentrated in four areas.

First, technological progress and its social implications constitute an important starting point for public discussion. Many commentators view autonomous driving as a major direction in the development of intelligent transportation and connect it to industrial upgrading, efficiency improvement, and the transformation of future mobility modes. This suggests that public

understanding of autonomous ride-hailing is not confined to a specific product, but is situated within a broader context of technological change and social transformation.

Second, safety and responsibility attribution remain at the center of discussion. Compared with conventional ride-hailing, the algorithmic decision-making, autonomous control, and emergency-handling capacity of driverless ride-hailing are more likely to trigger public concerns about accident risks, pedestrian protection, and compliance with traffic rules. Once accidents occur, the question of whether responsibility should be borne by the platform, the vehicle manufacturer, system developers, or other actors also becomes a major factor affecting user trust. Safety perception and institutional safeguards are therefore among the most sensitive dimensions in public evaluations of autonomous services.

Third, industry change and market competition are also frequent topics of concern. The public generally recognizes that the spread of autonomous driving may alter the structure of traditional driving jobs and reshape the competitive landscape of the taxi and ride-hailing sectors. On the one hand, autonomous driving may improve efficiency and reduce prices; on the other hand, the concentration of capital and technology behind it may generate worries about platform monopoly and market fairness.

Fourth, user experience and real application scenarios form the micro-level basis of public evaluation. Compared with abstract technological imagination, people are more concerned with how autonomous ride-hailing performs in real urban traffic situations, including ride comfort, operational stability, emergency-response capacity, and overall service convenience. This indicates that whether users ultimately accept autonomous services depends to a large extent on their practical perceptions and experience-based judgments in concrete application settings.

Overall, public concern regarding autonomous ride-hailing has developed into a multi-level cognitive structure that extends from macro-level technological development, to meso-level industry effects, to micro-level user experience. This preliminary analysis indicates that judgments about autonomous services are not built solely on technological novelty, but are jointly shaped by functional cognition, safety concerns, trust judgments, and real or anticipated experience.

3.3 Real-World Basis for Variable Selection and Model Construction

The preliminary analysis further suggests a clear progressive logic of cognition, trust, and behavior in public evaluations of autonomous ride-hailing. More specifically, users first form preliminary judgments about whether the service is useful and convenient; these cognitive evaluations then influence their trust in the system's safety, reliability, and controllability; finally, users form behavioral intentions regarding whether they are willing to try or continue using the service.

This real-world logic fits closely with TAM and trust theory. TAM emphasizes the foundational influence of perceived usefulness and perceived ease of use on behavioral intention, while trust theory highlights trust as the key psychological mechanism linking cognition and decision-making in high-risk and high-uncertainty technological settings. Autonomous ride-hailing is a typical example of such a context: users must judge not only whether the technology is efficient and convenient, but also whether it is safe, credible, and dependable.

Therefore, based on the preliminary analysis and the relevant theoretical foundation, this study takes perceived usefulness, perceived ease of use, trust, and usage intention as the core variables and constructs a cognition-trust-behavioral intention model. In this framework, perceived usefulness reflects users' judgments about the practical value of autonomous ride-hailing for improving travel efficiency and service experience; perceived ease of use captures users' perceptions of operational convenience and process simplicity; trust reflects users' subjective evaluations of system reliability, safety, and controllability; and usage intention represents users' future propensity to adopt autonomous ride-hailing services.

Accordingly, the significance of this chapter's preliminary analysis does not lie in an independent discussion of text-based methodology as such, but rather in extracting research questions from real-world discussions and user cognition, clarifying the basis for variable selection, and laying both theoretical and empirical foundations for the subsequent questionnaire design and structural-model testing.

4 Research Design and Empirical Analysis

4.1 Theoretical Hypotheses and Variable Measurement

4.1.1 Effects of External Contextual Factors on Perceived Ease of Use and Perceived Usefulness

TAM holds that perceived ease of use (PEOU) and perceived usefulness (PU) are the two core cognitive variables explaining user technology adoption. Existing studies show that when users perceive a technology as easy to operate and clear to interact with, they are more likely to form positive value judgments and stronger usage intention. This logic has been validated across a variety of technological settings, including autonomous driving, mobile payment, in-vehicle systems, and e-learning. At the same time, some studies indicate that the effect of perceived ease of use is context-dependent. Especially in high-uncertainty technological environments, users' judgments about technological value often depend not only on operational convenience, but also on institutional, market, and risk-related contextual conditions.

Given the practical characteristics of autonomous ride-hailing, this study introduces four categories of external contextual factors: Technological Progress and Social Transformation (TS), Benefit Distribution and Social Structure Adjustment (BS), Industrial Transformation and Market Competition (IM), and User Experience and Safety Risk Management (US). These factors are intended to identify key antecedents shaping users' cognitive evaluations.

First, TS captures the macro background of continued technological development and social diffusion of autonomous driving. Improvements in technological maturity, system reliability, policy support, and public understanding may all lower the psychological threshold for use and strengthen users' perceptions of technological value. Accordingly, the following hypotheses are proposed.

H1: Technological Progress and Social Transformation positively affects Perceived Ease of Use.

H2: Technological Progress and Social Transformation positively affects Perceived Usefulness.

Second, BS reflects public judgments about fairness, inclusiveness, and social consequences during technology diffusion. If users believe that the spread of autonomous driving can improve efficiency while also promoting social fairness, generating new employment opportunities, and preventing one-sided platform gains, they are more likely to form positive functional cognition and value evaluations. Accordingly:

H3: Benefit Distribution and Social Structure Adjustment positively affects Perceived Ease of Use.

H4: Benefit Distribution and Social Structure Adjustment positively affects Perceived Usefulness.

Third, IM reflects the restructuring effect of autonomous ride-hailing on the existing mobility-service landscape. Intensified market competition may drive platforms to improve service quality, optimize pricing, and strengthen functional innovation, thereby enhancing users' perception of convenience and practicality.

H5: Industrial Transformation and Market Competition positively affects Perceived Ease of Use.

H6: Industrial Transformation and Market Competition positively affects Perceived Usefulness.

Fourth, US is directly related to the process by which users form cognition in real or anticipated use contexts. Smooth riding experiences, clear interfaces, sound safety mechanisms, and visible risk-control capacity all help enhance users' perceptions of convenience and value in autonomous services.

H7: User Experience and Safety Risk Management positively affects Perceived Ease of Use.

H8: User Experience and Safety Risk Management positively affects Perceived Usefulness.

4.1.2 Effects of External Contextual Factors on Trust

In the autonomous ride-hailing context, trust is widely regarded as a key psychological mechanism shaping user adoption. Unlike conventional ride-hailing, where users often rely on judgments about individual drivers, driverless services require users to rely more heavily on trust in the platform, the algorithmic system, and the platform's safety-management capabilities. In high-uncertainty technological contexts, user trust is typically built on integrated judgments about system reliability, risk controllability, and the service provider's ability to assume responsibility.

Prior studies suggest that perceived risk is an important antecedent of trust formation. In autonomous driving, these risks mainly involve performance risk, safety risk, privacy risk, and legal-liability uncertainty. Trust therefore does not emerge in isolation; it is closely related to technological maturity, institutional fairness, market conditions, and concrete usage experience.

Within the context of this study, TS can reduce users' concerns about technological uncertainty and thereby strengthen judgments about reliability and controllability; BS reflects fairness and responsibility in the platform and technology-diffusion process and may enhance institutional trust; IM may improve service quality and professional image through competition, thereby reinforcing perceptions of trustworthiness; and US forms the practical basis of trust formation because system stability, safety, and transparency directly shape users' judgments of credibility. Accordingly, the following hypotheses are proposed.

H9: Technological Progress and Social Transformation positively affects Trust.

H10: Benefit Distribution and Social Structure Adjustment positively affects Trust.

H11: Industrial Transformation and Market Competition positively affects Trust.

H12: User Experience and Safety Risk Management positively affects Trust.

4.1.3 Perceptual Variables, Trust, and Usage Intention

Within TAM, usage intention (UI) is the core outcome variable used to measure user acceptance. A large number of studies show that perceived usefulness has a persistent and stable positive effect on usage intention. When users believe that a technology can improve efficiency, enhance experience, or generate concrete benefits, their willingness to adopt it usually increases significantly. This conclusion is also supported in the autonomous-driving literature.

By comparison, the effect of perceived ease of use on usage intention is less stable across contexts. On the one hand, PEOU can directly increase adoption tendency by reducing learning costs and operational difficulty. On the other hand, in complex new-technology settings, its effect may be realized more indirectly through improving PU. Accordingly, in the autonomous ride-hailing context, PEOU is likely to have both direct and indirect effects.

In addition to cognitive variables, trust is also an important determinant of users' usage intention. Because autonomous driving involves relatively high uncertainty and potential risk, users often rely on trust to compensate for limited experience and incomplete cognition when making usage decisions. Trust is therefore not only an independent explanatory variable, but also a key bridge linking cognitive evaluations and behavioral intention. Based on this reasoning, the following hypotheses are proposed.

H13: Perceived Usefulness positively affects Usage Intention.

H14: Perceived Ease of Use positively affects Usage Intention.

H15: Trust positively affects Usage Intention.

4.1.4 Mediation Hypotheses

Given that autonomous ride-hailing is a high-uncertainty technological service, users are unlikely to form final adoption intention directly on the basis of external contextual judgments alone. Instead, they are more likely to reach behavioral decisions through cognitive evaluation and trust mechanisms. This study therefore further examines the mediating roles of perceived ease of use, perceived usefulness, and trust between external contextual factors and usage intention.

More specifically, the four categories of external variables - TS, BS, IM, and US - may indirectly influence usage intention through PEOU, PU, and trust. The following mediation hypotheses are proposed.

H16a: Perceived Ease of Use mediates the relationship between Technological Progress and Social Transformation and Usage Intention.

H16b: Perceived Usefulness mediates the relationship between Technological Progress and Social Transformation and Usage Intention.

H16c: Trust mediates the relationship between Technological Progress and Social Transformation and Usage Intention.

H17a: Perceived Ease of Use mediates the relationship between Benefit Distribution and Social Structure Adjustment and Usage Intention.

H17b: Perceived Usefulness mediates the relationship between Benefit Distribution and Social Structure Adjustment and Usage Intention.

H17c: Trust mediates the relationship between Benefit Distribution and Social Structure Adjustment and Usage Intention.

H18a: Perceived Ease of Use mediates the relationship between Industrial Transformation and Market Competition and

Usage Intention.

H18b: Perceived Usefulness mediates the relationship between Industrial Transformation and Market Competition and Usage Intention.

H18c: Trust mediates the relationship between Industrial Transformation and Market Competition and Usage Intention.

H19a: Perceived Ease of Use mediates the relationship between User Experience and Safety Risk Management and Usage Intention.

H19b: Perceived Usefulness mediates the relationship between User Experience and Safety Risk Management and Usage Intention.

H19c: Trust mediates the relationship between User Experience and Safety Risk Management and Usage Intention.

Table 4-1. Summary of Research Hypotheses

Hypothesis	Path	Hypothesis	Path
H1	TS → PEOU	H9	TS → Trust
H2	TS → PU	H10	BS → Trust
H3	BS → PEOU	H11	IM → Trust
H4	BS → PU	H12	US → Trust
H5	IM → PEOU	H13	PU → UI
H6	IM → PU	H14	PEOU → UI
H7	US → PEOU	H15	Trust → UI
H8	US → PU	H16a–H19c	Indirect effects of PEOU, PU, and Trust between external variables and UI

4.2 Questionnaire Design, Data Collection, and Sample Characteristics

4.2.1 Questionnaire Design and Scale Sources

Based on established scales, this study made moderate contextual revisions to the measurement items to fit the autonomous ride-hailing setting. More specifically, the PEOU scale mainly draws on Liu et al. (2024), the PU scale on Jaszberenyi et al. (2022), the trust scale on Du et al. (2021), and the usage-intention scale on Davis (1989). The four external contextual variables were developed on the basis of preliminary literature review, contextual analysis, and the theoretical framework so as to ensure content validity in the autonomous ride-hailing setting.

To ensure linguistic equivalence and content validity, all items were processed through a translation-back-translation procedure. Two researchers with academic training in English first completed the initial translation, after which two domain experts reviewed and revised the item wording to improve accuracy, clarity, and contextual fit. All measurement items were rated on a five-point Likert scale.

Table 4-2. Measurement Items and Sources

Construct	Item Code	Measurement Item	Source
Technological Progress and Social Transformation (TS)	TS1	I believe autonomous driving significantly boosts societal productivity.	Developed for this study
	TS2	I see autonomous driving as a key indicator of technological progress.	Developed for this study
	TS3	With autonomous driving, related jobs will require new skills and roles.	Developed for this study
	TS4	The popularization of autonomous driving will reshape social perceptions and occupational structures.	Developed for this study
	TS5	Autonomous driving will enhance industrial efficiency while reshaping human-machine relations.	Developed for this study
Benefit Distribution and Social Structure Adjustment (BS)	BS1	I believe autonomous driving can improve efficiency but may widen income gaps.	Developed for this study
	BS2	The adoption of autonomous driving will create new jobs and service opportunities.	Developed for this study
	BS3	I trust autonomous driving platforms to operate fairly without harming smaller	Developed for this study

		operators.	
	BS4	The spread of autonomous driving will reshape social structures by changing control over productive resources.	Developed for this study
	IM1	Autonomous driving will reshape competition between traditional taxis and ride-hailing services.	Developed for this study
Industrial Transformation and Market Competition (IM)	IM2	Autonomous driving will prompt platforms to compete through more flexible pricing strategies.	Developed for this study
	IM3	Autonomous driving platforms could reshape market dynamics, affecting pricing and service choices.	Developed for this study
	IM4	Autonomous driving will diversify transportation options and promote a more efficient mobility market.	Developed for this study
	US1	I find autonomous driving rides convenient and easy to use.	Developed for this study
User Experience and Safety Risk Management (US)	US2	The safety measures in autonomous vehicles make me feel protected.	Developed for this study
	US3	I trust the platform to manage vehicle algorithms and monitoring systems properly.	Developed for this study
	US4	Riding in an autonomous vehicle gives me a sense of novelty and technological advancement.	Developed for this study
Perceived Ease of Use (PEOU)	PEOU1	It is easy to use an autonomous taxi.	Liu et al. (2024)
	PEOU2	Using an autonomous taxi does not require much mental effort.	Liu et al. (2024)
	PEOU3	Interacting with an autonomous taxi is straightforward and makes sense.	Liu et al. (2024)
Perceived Usefulness (PU)	PU1	Autonomous driving allows me to relax and focus on other things during the trip.	Jászberényi et al. (2022)
	PU2	Autonomous driving reduces my need to constantly monitor the road environment.	Jászberényi et al. (2022)
	PU3	Autonomous driving ensures safe arrival without relying on a driver.	Jászberényi et al. (2022)
Trust	T1	Self-driving cars are reliable.	Du et al. (2021)
	T2	Self-driving cars are dependable.	Du et al. (2021)
	T3	Overall, I can trust self-driving cars.	Du et al. (2021)
Usage Intention (UI)	UI1	I expect to try using autonomous taxis in the future.	Davis (1989)
	UI2	If given the opportunity, I am willing to use an autonomous taxi now.	Davis (1989)
	UI3	If the price is reasonable, I intend to use autonomous taxis in the future.	Davis (1989)

4.2.2 Data Collection and Sample Screening

The survey targeted internet users with a certain degree of familiarity with Apollo Go, with the aim of examining the mechanisms linking cognitive evaluation, trust formation, and usage intention among potential users of autonomous ride-hailing. The questionnaire was distributed online through the Wenjuanxing platform in August 2025, and a total of 667 responses were collected.

To ensure sample quality, the raw data were screened rigorously. Questionnaires completed in an excessively short time were removed to exclude likely careless responses; cases with highly uniform response patterns or obvious logical contradictions were excluded; and respondents lacking basic knowledge of the research object were also removed. After screening, 633 valid questionnaires were retained. This sample size satisfies the basic requirements of PLS path modeling and provides a sufficiently robust foundation for subsequent reliability, validity, and path analyses.

4.2.3 Descriptive Statistics of the Sample

The sample structure shows that male respondents accounted for 51.3% and female respondents for 48.7%, indicating a relatively balanced gender distribution. In terms of age, the largest proportion was in the 30-34 group, followed by respondents aged 36 and above, suggesting that the sample was concentrated in middle-aged and younger adult groups with relatively strong technology-acceptance capacity and purchasing power. With respect to income, the sample covered several income levels, with middle-income groups accounting for a larger share, which is helpful for identifying the potential role of economic factors in adoption behavior. In terms of education, most respondents held a bachelor's degree or above, indicating relatively strong

technological understanding and information-processing ability across the sample.

Table 4-3. Descriptive Statistics of the Sample

Category	Group	Frequency	Percentage (%)
Gender	Male	325	51.3
	Female	308	48.7
Age	Less than 24	60	9.5
	25–29	60	9.5
	30–34	326	51.5
	36 and above	187	29.5
	Less than 7,000	51	8.1
Annual income (USD)	7,001–10,000	167	26.4
	10,001–13,500	297	46.9
	13,501–17,000	81	12.8
	Above 17,001	37	5.8
	Associate degree and below	66	10.4
Education	Bachelor's degree	435	68.7
	Postgraduate and above	132	20.9

Overall, the sample covers the major potential user groups for autonomous ride-hailing while also retaining a degree of demographic diversity, thereby providing a sound empirical basis for the subsequent analysis.

5 Empirical Results

5.1 Measurement Model Assessment

5.1.1 Reliability and Convergent Validity

Before testing the structural model, the study first evaluated the reliability and validity of the measurement model to ensure that each latent construct could be reflected stably and accurately by its corresponding indicators. The results show that the standardized factor loadings of all items exceed 0.70, indicating good explanatory ability of the items for their respective constructs. In addition, the Cronbach's alpha and composite reliability (CR) values for all latent variables are above 0.70, indicating good internal consistency.

Regarding convergent validity, the average variance extracted (AVE) values of all constructs exceed 0.50, suggesting that the latent variables explain a substantial share of variance in their indicators and meet the criterion for convergent validity. Moreover, all variance inflation factor (VIF) values are below 3.30, indicating that no serious multicollinearity problem exists in the model. Overall, the measurement model meets the statistical standards for reliability and convergent validity and is therefore suitable for subsequent analysis.

5.1.2 Discriminant Validity

Discriminant validity was assessed using the Fornell-Larcker criterion and the heterotrait-monotrait ratio (HTMT). According to the Fornell-Larcker criterion, the square root of each construct's AVE is greater than its correlations with other constructs, indicating good discriminant distinction among constructs.

At the same time, all HTMT values are below the recommended threshold of 0.90, which further confirms adequate discriminant validity. Taken together, both the traditional Fornell-Larcker approach and the more stringent HTMT test indicate that the latent constructs in this study are clearly distinguishable, demonstrating a sound foundation of construct validity for the measurement model.

5.1.3 Model Fit, Explanatory Power, and Effect Sizes

For overall model fit, the study uses the standardized root mean square residual (SRMR) as the evaluation indicator. The SRMR value is 0.055, which is substantially lower than the recommended threshold of 0.09, indicating good overall model fit

and suitability for path analysis.

In terms of explanatory power, the adjusted R^2 values show that the explained variances of perceived ease of use (PEOU), perceived usefulness (PU), trust, and usage intention (UI) are 0.178, 0.147, 0.167, and 0.105, respectively. Although these values fall within a weak-to-moderate range in a strict statistical sense, such levels remain acceptable in technology-acceptance and user-behavior research, especially in models involving complex social-cognitive and contextual variables. This suggests that the model can explain the formation mechanism of user cognition, trust, and behavioral intention toward autonomous ride-hailing to a meaningful extent.

The study further evaluates local effect sizes using f^2 . Most paths exhibit small effects, while a few are below 0.02. It should be noted that f^2 primarily reflects the marginal explanatory contribution of a single path and does not imply an absence of theoretical significance. Given the strong contextual specificity of the external variables introduced in this study and the focus on testing their mechanism in the autonomous ride-hailing setting, even paths with relatively small effect sizes retain interpretive value so long as their coefficients are statistically significant and consistent with theoretical expectations.

Table 5-1. Measurement Model Results

Construct	Item	Loading	VIF	Cronbach's Alpha	CR	AVE
TS	TS1	0.817	1.968	0.877	0.911	0.671
	TS2	0.803	1.927			
	TS3	0.825	1.991			
	TS4	0.833	2.218			
	TS5	0.817	1.981			
BS	BS1	0.799	1.757	0.838	0.892	0.673
	BS2	0.822	1.763			
	BS3	0.828	1.948			
	BS4	0.831	1.835			
IM	IM1	0.824	1.82	0.845	0.896	0.682
	IM2	0.811	1.761			
	IM3	0.824	1.942			
	IM4	0.844	1.98			
US	US1	0.814	1.704	0.83	0.887	0.661
	US2	0.792	1.73			
	US3	0.823	1.872			
	US4	0.824	1.753			
PU	PU1	0.842	1.718	0.795	0.88	0.709
	PU2	0.826	1.635			
	PU3	0.859	1.716			
PEOU	PEOU1	0.84	1.716	0.798	0.881	0.712
	PEOU2	0.856	1.704			
	PEOU3	0.836	1.689			
Trust	Trust1	0.847	1.628	0.777	0.87	0.691
	Trust2	0.827	1.654			
	Trust3	0.819	1.535			
UI	UI1	0.832	1.662	0.796	0.878	0.707
	UI2	0.869	1.655			
	UI3	0.82	1.766			

Table 5-2. Discriminant Validity Test - Panel A. Fornell-Larcker Criterion

	BS	IM	PEOU	PU	TS	Trust	UI	US
BS	0.82							
IM	0.237	0.826						
PEOU	0.262	0.309	0.844					

PU	0.284	0.245	0.274	0.842				
TS	0.17	0.291	0.279	0.245	0.819			
Trust	0.163	0.296	0.308	0.294	0.285	0.831		
UI	0.209	0.302	0.23	0.275	0.195	0.202	0.841	
US	0.235	0.263	0.268	0.236	0.237	0.301	0.169	0.813

Table 5-2. Discriminant Validity Test - Panel B. HTMT Ratio

	BS	IM	PEOU	PU	TS	Trust	UI	US
BS								
IM	0.279							
PEOU	0.317	0.374						
PU	0.343	0.298	0.342					
TS	0.197	0.337	0.332	0.292				
Trust	0.201	0.364	0.388	0.372	0.343			
UI	0.255	0.368	0.28	0.337	0.236	0.252		
US	0.279	0.313	0.326	0.288	0.276	0.373	0.209	

Table 5-3. Model Explanatory Power and Effect Sizes

Endogenous Variable / Path	Value	Endogenous Variable / Path	Value
Adjusted R ² (PEOU)	0.178	BS → Trust	0.002
Adjusted R ² (PU)	0.147	IM → PEOU	0.036
Adjusted R ² (Trust)	0.167	IM → PU	0.015
Adjusted R ² (UI)	0.105	IM → Trust	0.034
SRMR	0.055	US → PEOU	0.022
TS → PEOU	0.029	US → PU	0.015
TS → PU	0.022	US → Trust	0.043
TS → Trust	0.033	PEOU → UI	0.02
BS → PEOU	0.027	PU → UI	0.042
BS → PU	0.043	Trust → UI	0.009

5.2 Structural Model Assessment

5.2.1 Direct Effects

Table 5-4. Direct Effects Test Results

Hypothesis	Path	β	Mean	SD	t	p	Decision
H1	TS → PEOU	0.164	0.165	0.039	4.238	0	Supported
H2	TS → PU	0.146	0.147	0.037	3.998	0	Supported
H3	BS → PEOU	0.156	0.157	0.038	4.142	0	Supported
H4	BS → PU	0.201	0.202	0.037	5.427	0	Supported
H5	IM → PEOU	0.187	0.188	0.039	4.764	0	Supported
H6	IM → PU	0.123	0.124	0.041	3.026	0	Supported
H7	US → PEOU	0.144	0.144	0.038	3.739	0	Supported
H8	US → PU	0.122	0.123	0.038	3.192	0	Supported
H9	TS → Trust	0.177	0.178	0.037	4.808	0	Supported
H10	BS → Trust	0.043	0.043	0.038	1.122	0.26	Unsupported
H11	IM → Trust	0.181	0.182	0.038	4.771	0	Supported
H12	US → Trust	0.201	0.202	0.037	5.393	0	Supported
H13	PEOU → UI	0.144	0.144	0.04	3.582	0	Supported

H14	PU → UI	0.207	0.208	0.038	5.506	0	Supported
H15	Trust → UI	0.096	0.097	0.042	2.304	0.02	Supported

The direct paths in the structural model were tested using a nonparametric bootstrap procedure with 5,000 resamples. The results show that Technological Progress and Social Transformation (TS) has significant positive effects on both PEOU and PU, supporting H1 and H2. Benefit Distribution and Social Structure Adjustment (BS) also has significant positive effects on PEOU and PU, supporting H3 and H4. Industrial Transformation and Market Competition (IM) positively affects both PEOU and PU, supporting H5 and H6. User Experience and Safety Risk Management (US) likewise significantly enhances both PEOU and PU, supporting H7 and H8.

With respect to trust formation, TS, IM, and US all have significant positive effects on trust, supporting H9, H11, and H12. However, the effect of BS on trust is not statistically significant, so H10 is not supported. This suggests that, compared with distributional fairness and social-structure concerns, technological maturity, competitive environment, and actual experience are more capable of directly shaping users' trust judgments regarding autonomous ride-hailing systems.

At the outcome level, PEOU, PU, and trust all exert significant positive effects on UI, supporting H13, H14, and H15. Notably, the path coefficient from PU to UI is larger than those from PEOU and trust, indicating that in the autonomous ride-hailing context, users are more concerned with practical utility and functional value than with operational convenience or generalized trust alone. Overall, all direct-effect hypotheses are supported except H10. This indicates that external contextual variables can promote the formation of adoption intention by shaping users' functional cognition and trust judgments.

5.2.2 Mediation Effects

Table 5-5. Mediation Effects Test Results

Hypothesis	Indirect Path	β	Mean	SD	t	p	Decision
H16a	TS → PEOU → UI	0.02	0.024	0.01	2.74	0.01	Supported
H16b	TS → PU → UI	0.03	0.031	0.01	3.21	0	Supported
H16c	TS → Trust → UI	0.02	0.017	0.01	2.05	0.04	Supported
H17a	BS → PEOU → UI	0.02	0.023	0.01	2.58	0.01	Supported
H17b	BS → PU → UI	0.04	0.042	0.01	3.69	0	Supported
H17c	BS → Trust → UI	0	0.004	0	0.94	0.35	Unsupported
H18a	IM → PEOU → UI	0.03	0.027	0.01	2.64	0.01	Supported
H18b	IM → PU → UI	0.03	0.026	0.01	2.48	0.01	Supported
H18c	IM → Trust → UI	0.02	0.018	0.01	1.9	0.06	Unsupported
H19a	US → PEOU → UI	0.02	0.021	0.01	2.48	0.01	Supported
H19b	US → PU → UI	0.03	0.026	0.01	2.68	0.01	Supported
H19c	US → Trust → UI	0.02	0.02	0.01	2.12	0.03	Supported

*Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Following the recommendation of Preacher and Hayes (2008), the study further uses bootstrap analysis to test the mediating roles of perceived ease of use, perceived usefulness, and trust between external contextual variables and usage intention.

The results show that TS has significant indirect effects on UI through PEOU, PU, and trust, supporting H16a, H16b, and H16c. This indicates that technological progress not only improves users' functional evaluations of autonomous services, but also strengthens their trust, thereby indirectly promoting adoption willingness.

For BS, significant indirect effects on UI are found through PEOU and PU, supporting H17a and H17b, whereas the indirect effect through trust is not significant, so H17c is not supported. This suggests that distribution-related factors mainly promote usage intention by shaping cognition regarding convenience and practicality, but do not effectively translate into deeper trust mechanisms.

IM likewise shows significant mediation through PEOU and PU, supporting H18a and H18b, whereas the indirect path through trust is not significant and H18c is not supported. This implies that competitive market conditions more readily improve users' cognitive evaluations through service quality, pricing mechanisms, and functional optimization, while their transmission to trust is relatively limited.

US exhibits the most complete mediation pattern: it has significant positive indirect effects on UI through PEOU, PU, and trust, supporting H19a, H19b, and H19c. This suggests that good riding experience and effective risk management not only enhance users' evaluations of convenience and practicality, but also strengthen trust in system reliability, thereby effectively promoting adoption behavior.

Taken together, PEOU and PU are the most stable mediators through which the four categories of external contextual variables influence usage intention, whereas the mediating role of trust is more context-dependent and becomes significant only along the paths from TS and US. This indicates that in the autonomous ride-hailing context, trust does not transmit the influence of all external factors equally; rather, it is more readily driven by technological maturity and actual or anticipated usage experience.

5.2.3 Summary of Hypothesis Testing Results

Overall, most of the hypotheses proposed in this study are supported. In terms of direct effects, only the path from Benefit Distribution and Social Structure Adjustment to Trust fails to pass the test. In terms of mediation, only the BS → Trust → UI path and the IM → Trust → UI path are not supported. All other hypotheses are statistically significant and consistent with theoretical expectations.

These results provide broad support for the theoretical framework of external contextual factors, cognition and trust, and usage intention developed in this study. They indicate that user adoption of autonomous ride-hailing is not determined by a single factor, but is jointly shaped by the technological environment, market environment, social perceptions, user experience, and trust mechanisms.

6 Key Findings and Mechanism Interpretation

6.1 External Contextual Factors Shape the Cognitive Basis of User Adoption

The first key finding of this study is that all four categories of external contextual factors - TS, BS, IM, and US - significantly enhance users' perceived ease of use and perceived usefulness. This suggests that in the autonomous ride-hailing context, users' acceptance of the technology is not built solely on product functions, but is deeply influenced by the broader social setting, institutional environment, market structure, and usage experience in which the technology is embedded.

The underlying mechanism is that users' cognition of autonomous driving is not formed in isolation, but is embedded in a broader sociotechnical system. TS significantly improves PEOU and PU because users tend to view autonomous driving as a marker of technological progress, industrial upgrading, and the evolution of future mobility. When a technology is endowed with stronger social meaning and developmental legitimacy, users not only find it easier to understand its operating logic, but are also more likely to recognize its practical value. In other words, the social visibility and cultural legitimacy of a technology enhance positive evaluations of its convenience and utility through the dual processes of meaning attribution and value internalization.

The significant effect of BS on PU and PEOU indicates that when evaluating autonomous-driving technology, users do not merely perform individual-level utility calculations; they also consider whether the social contract behind the technology is reasonable. When the technology is perceived as contributing to social efficiency, fair competition, or more inclusive service provision, users are more likely to assign it a higher degree of social legitimacy. This means that technology acceptance partly contains an expectation of a better social order, and that such normative judgment is projected onto cognition of whether the technology is worth using and easy to use.

The positive effect of IM further suggests that users are sensitive to the potential of autonomous driving to reshape the structure of mobility markets. When users expect autonomous driving to optimize pricing mechanisms, stimulate service

innovation, and diversify travel options, they are more likely to judge the technology as worthy of adoption. In this sense, market competition is not merely a background variable in technology diffusion; by changing user expectations and strengthening perceptions of comparative advantage, it is translated into stronger subjective perceptions of convenience and utility.

Finally, the significant promoting effect of US on PU and PEOU indicates that acceptance of autonomous driving is, in essence, also an experience-driven process. When users perceive smooth procedures, timely feedback, sound safety protection, and transparent systems in actual or anticipated interaction, they gain a stronger sense of control and predictability, which in turn raises their evaluations of ease of use and usefulness. Notably, the novelty and technological aura associated with autonomous driving may also trigger positive emotional responses, leading users to recognize technological value not only functionally, but also affectively.

Overall, this finding suggests that PU and PEOU in autonomous-driving contexts are not purely individual cognitive variables; rather, they are jointly shaped by social meaning, institutional environment, market expectations, and experiential feedback. To some extent, this extends the boundary of traditional TAM and indicates that in autonomous-system contexts, technology acceptance has become a composite psychological process combining functional judgment, social judgment, and affective identification.

6.2 Trust Formation Is Driven by Technology, Experience, and Competition Rather Than by Institutions Alone

The second key finding of this study is that TS, IM, and US all significantly enhance user trust, whereas BS does not have a significant effect on trust. This indicates that trust formation in the autonomous ride-hailing context is selective: users are more likely to trust technologies and experiences that are visible, tangible, and verifiable, and are less likely to establish direct trust in a platform or system solely on the basis of abstract perceptions of institutional fairness.

Mechanistically, TS promotes trust because it strengthens users' symbolic identification with autonomous driving. Users often view autonomous driving as a manifestation of technological maturity, social progress, and future development. This kind of anticipatory trust does not depend entirely on actual usage experience, but is partly formed even before direct contact with the technology. Once autonomous driving is understood as a symbol of social progress, users are more inclined to assume that it is advanced, reliable, and trustworthy.

The significant effect of IM on trust reveals another structural mechanism of trust generation. Active market competition is seen by users as a form of external supervision and balancing force. It implies that platforms cannot easily form monopolies, that users retain greater choice, and that market operations are more transparent. Under such conditions, users' trust in a platform is not derived solely from the platform itself, but also from the institutional assurance embedded in the competitive environment. Competition therefore has not only an economic function, but also a structural trust-generating function.

By contrast, the non-significant effect of BS on trust shows that fairness in distribution and macro-level social restructuring remain largely at the level of rational cognition and do not readily translate into trust in a specific technological system. Users may recognize that autonomous driving could change income distribution, occupational structure, and resource allocation, but such macro-structural cognition does not automatically become the micro-level judgment that the system itself is trustworthy. In forming trust, users care more about whether the system is safe, reliable, and controllable than whether it is institutionally fair. This implies that trust in autonomous-driving contexts is more technology-oriented than institution-oriented.

The significant promoting effect of US on trust further shows that trust is ultimately rooted in embodied experience. Only when users feel that the system is transparent, monitored, controllable in risk, and comfortable in actual or anticipated use do they build durable trust. In autonomous-system settings, trust is not generated by abstract promises alone, but accumulated through repeatedly validated feelings of safety, control, and protection.

Overall, the study indicates that trust formation in autonomous ride-hailing follows a triple logic driven by technological perception, competitive structure, and experiential feedback. This finding enriches the applicability of the TAM-trust framework in autonomous-system contexts and suggests that platforms seeking to build stable user trust must prioritize visible technological reliability, transparent operation, and sustained positive experience rather than relying only on macro-level

institutional narratives.

6.3 The Mediation Mechanism Exhibits a Dual-Path Structure: Cognitive Mediation Is More Stable, Whereas Trust Mediation Is More Context-Dependent

The third key finding of this study is that perceived ease of use and perceived usefulness play stable mediating roles between all four categories of external contextual variables and usage intention, whereas the mediating role of trust varies clearly across contexts. More specifically, TS and US influence usage intention jointly through PEOU, PU, and trust, whereas BS and IM mainly operate through PEOU and PU, with the mediating paths through trust remaining non-significant.

This result indicates that the adoption mechanism of autonomous ride-hailing is not a single linear process, but involves two parallel psychological paths: one is a rational-cognitive path centered on convenience and utility, and the other is an affective-trust path centered on understanding, trusting, and adopting the technology.

TS can comprehensively affect usage intention through all three paths because it acts simultaneously at both rational and affective levels. When users see autonomous driving as a symbol of technological progress and industrial upgrading, they more readily perceive its convenience and utility. At the same time, this symbolic meaning may evoke social identification and technological pride, thereby reducing uncertainty and enhancing trust. In this sense, technological progress strengthens not only the judgment that the technology is useful and easy to use, but also the willingness to trust it, thus generating a complete adoption chain.

US likewise operates through all three paths, indicating that experiential variables in the autonomous-driving context serve both as cognitive bridges and as affective catalysts. When users perceive the system as safe, reliable, transparent in interaction, and comfortable in use, they not only judge it to be easy and worthwhile to use, but also develop affective trust in the system. This means that in autonomous systems, experience is not a peripheral supplementary variable; rather, it is a key mechanism through which technological attributes are converted into behavioral intention.

By contrast, BS and IM mainly influence usage intention through PEOU and PU and fail to form stable mediation through trust. This indicates that although these variables can shape rational judgments and lead users to recognize the potential efficiency gains, market optimization, or social value of autonomous driving, such cognitive improvement does not automatically become affective trust. Users may acknowledge that the technology has value without necessarily trusting it. This reveals an important feature of the autonomous-driving adoption process: institutional and competitive factors primarily shape rational evaluation, whereas the generation of trust still depends more strongly on direct experience and technological verifiability.

Overall, the study reveals a dual-path parallel mechanism in user adoption of autonomous ride-hailing: TS and US are more likely to create a complete understanding-trust-adoption path, whereas BS and IM mainly generate cognitive mediation through utility, convenience, and adoption. This finding not only extends the explanatory power of the TAM-trust model in autonomous-system contexts, but also suggests that future platform governance should distinguish between two logics of adoption promotion. For institutional and competitive variables, the priority is to strengthen their transmission to user value perception; for technological progress and user-experience variables, the key is to transform them into visible trust signals so as to enable a transition from rational acceptance to affective trust.

7 Conclusion and Future Prospects

Drawing on the Technology Acceptance Model and trust theory, this study constructs and validates an integrated analytical framework linking external contextual factors, cognitive evaluation, trust, and usage intention, and systematically reveals the multi-layered mechanism underlying user adoption of autonomous ride-hailing. The findings show that adoption behavior is not determined by technological performance alone. Rather, it is formed gradually through cognitive-evaluation mechanisms such as perceived usefulness and perceived ease of use under the joint influence of social context, market structure, and usage experience, and is then deepened into a more stable behavioral intention through trust. In high-uncertainty technological settings,

trust serves as a key mediating mechanism linking cognitive judgment and final adoption behavior.

More specifically, external factors such as technological progress and social transformation, benefit distribution and social structure adjustment, industrial transformation and market competition, and user experience and safety risk management all significantly influence users' cognitive judgments regarding the convenience and practical value of the technology. Some of these factors - especially technological progress and user experience - further stimulate user trust and thereby create deeper behavioral-driving mechanisms. This indicates that in autonomous-driving contexts, user adoption is not merely a rational evaluation of technical functionality, but also a comprehensive response to the technology's social meaning, operating environment, and interactional experience.

Compared with traditional technology-acceptance research, which mainly focuses on functional cognition at the individual level, this study further demonstrates that the acceptance of autonomous-driving technology exhibits a dual-mechanism structure in which cognitive and trust paths operate in parallel. On the one hand, users form a rational basis for adoption by evaluating usefulness and ease of use; on the other hand, they gradually build affective trust through perceptions of system safety, reliability, and controllability, which in turn strengthens behavioral intention. This multi-layered cognition-trust-behavior linkage highlights the critical role of social meaning, competitive structure, and experiential feedback in technology adoption.

On this basis, the study theoretically extends the explanatory boundary of TAM in autonomous-system contexts by bringing social-context variables and trust mechanisms into a unified analytical framework. At the practical level, the results indicate that platforms should improve algorithmic transparency, optimize user experience and safety-feedback mechanisms, and strengthen market competition and public communication in order to facilitate a transition from rational acceptance to affective trust. At the policy level, a trust-governance system coordinating technological transparency, institutional safeguards, and social communication is needed to support the long-term social acceptance of autonomous-driving technologies.

Several limitations should also be noted. For example, cross-sectional data make it difficult to capture the dynamic evolution of trust and adoption over time, and the research context and sample source constrain the external generalizability of the conclusions to some extent. Future research can combine longitudinal data, multi-source behavioral information, and cross-group comparisons to further test the applicability of the model across different institutional environments and application settings, thereby advancing research on autonomous-technology acceptance from an individual-rationality model toward a social-trust-system model.

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